The Laryngoscope
© 2012 The American Laryngological,
Rhinological and Otological Society, Inc.

Quantitative Electromyography Improves Prediction in Vocal Fold Paralysis

Libby J. Smith, DO, FAOCO; Clark A. Rosen, MD, FACS; Christian Niyonkuru, MS;
Michael C. Munin, MD

Objectives: Quantitative laryngeal electromyography (LEMG) using turns analysis can differentiate acute vocal fold paralysis from normal controls. The objective of this study is to determine if using both traditional qualitative LEMG measurements in addition to turns analysis improves prognostic accuracy in patients with acute vocal fold paralysis who demonstrate voluntary motor activity.

Study Design: Retrospective review of LEMG data (qualitative and quantitative) and charts of patients with vocal fold paralysis on flexible laryngoscopy, recurrent laryngeal neuropathy, and varying degrees of motor unit recruitment.

Methods: Laryngeal EMG using a standardized protocol involving qualitative (evaluation of recruitment, motor unit configuration, detection of fibrillations, synkinesis) and quantitative (turns analysis) measurements was performed. Prognosis was correlated with vocal fold motion recovery status (minimum of 6 months following onset) using positive and negative predictive values (PPV, NPV).

Results: Twenty-three patients underwent LEMG for acute recurrent laryngeal neuropathy. All four patients with excellent LEMG prognosis recovered motion, whereas 17/19 patients with fair/poor LEMG prognosis were without motion at least 6 months following onset, resulting in a 100% PPV and 89.5% NPV.

Conclusions: Integrating both qualitative and quantitative LEMG data improves prognostic accuracy in vocal fold paralysis patients who demonstrate voluntary motor unit activity.

Key Words: Quantitative laryngeal electromyography, LEMG, turns analysis, vocal fold paralysis, vocal cord paralysis, predicting prognosis.

Level of evidence: Level 4, case series

***Laryngoscope,* 122:854–859, 2012**

INTRODUCTION

Laryngeal electromyography (LEMG) is used to evaluate the type and degree of neurogenic injury of acute vocal fold paralysis. It is often used to estimate the prognosis for recovery of vocal fold motion, which helps guide further treatment. When patients have no motor unit recruitment and only spontaneous electrical activity in the form of fibrillations and positive waves, prognosis for motion recovery is usually poor. However, when some degree of motor recruitment is present, the process of determining an accurate prognosis for motion recovery is challenging because the amount and nature of motor unit recovery necessary to restore vocal fold (VF) motion is unknown. Evaluation of traditional LEMG measures, including recruitment, motor unit configuration, and detection of fibrillations, resulted in a positive predictive value (PPV) of 80% and negative predictive value (NPV) between 67 and 71%.[1,2] Once synkinesis data was incorporated into the prognosis determination algorithm, the PPV increased from 67 to 76% and the NPV from 53 to 100%.[3]

Despite the advancements and improvements in analyzing LEMG, one of the limitations has been its qualitative nature. The quality of the study is dependent upon the skill of the electromyographer doing the assessment. In addition, the use of different subjective grading scales, such as motor recruitment, cause confusion when comparing data from different centers. Quantitative LEMG analysis in the form of turns and amplitude analysis was previously evaluated for its utility in identifying patients with acute VF paralysis from normal controls.[4] Turns analysis quantitatively measures the interference pattern, which describes motor unit (MU) recruitment and is dependent upon the configuration of individual motor unit potentials (MUPs) (e.g., amplitude, duration, polyphasia), MU firing rates, and the number of active units discharging within the sampled field. As more MUs are activated, there will be more "turns." A "turn" is defined by a change in signal direction at the peak of a MU signal, with each successive turn separated by

From the University of Pittsburgh Voice Center (L.J.S., C.A.R., M.C.M.), Department of Otolaryngology—Head and Neck Surgery, University of Pittsburgh, Pittsburgh, Pennslyvania, U.S.A.; and Department of Physical Medicine and Rehabilitation (C.N., M.C.M.), University of Pittsburgh, Pittsburgh, Pennsylvania, U.S.A.

Editor's Note: This Manuscript was accepted for publication April 27, 2011.

The authors have no financial disclosures for this article.
The authors have no conflicts of interest to disclose.

Send correspondence to: Dr. Libby J. Smith, UPMC Mercy, 1400 Locust Street, Building B, Suite 11500, Pittsburgh, PA 15219. E-mail: smithlj2@upmc.edu

DOI: 10.1002/lary.21884

greater than 100 μV to exclude low-amplitude peaks generated by noise and electrical interference. Statham et al.[4] found that patients with acute VF paralysis had a mean of less than 400 turns/sec, whereas normal controls had a mean of greater than 400 turns/sec.

Our objective of this study was to determine the PPVs and NPVs for motion recovery after acute VF paralysis when some voluntary LEMG signal was present using both qualitative and quantitative LEMG findings to determine if accuracy could be improved.

MATERIALS AND METHODS

Study Design and Subjects

We retrospectively reviewed 89 consecutive LEMGs performed at the University of Pittsburgh Voice Center between June 2009 and September 2010. University of Pittsburgh institutional review board approval was obtained prior to collecting data. Patients included for analysis had clinical VF paralysis on flexible laryngoscopy, which was confirmed by LEMG. Paralysis had to be acute, between 3 and 27 weeks. Only patients with known VF mobility status at 6 months were included in the study because reinnervation is expected by that time.[5] Patients with known tumor invasion or disruption of the recurrent laryngeal nerve (RLN), VF paresis, cricoarytenoid joint pathology, or a history of systemic neurologic disease were excluded.

Electromyography and Analysis

The laryngeal electromyogram was performed by a board-certified otolaryngologist (L.J.S., C.A.R.), who was blinded to the patient's clinical diagnosis or side of suspected pathology. A board-certified electrodiagnostic medicine physician (M.C.M) was also present to interpret the data, knowing only the time frame of injury and past medical and surgical history, but not the side of clinical VF paralysis or laryngeal exam findings. A 37-mm concentric needle electrode was utilized in conjunction with a ground electrode placed over the clavicle. A Synergy EMG machine (Cardinal Health, Dublin, OH) was used, recording motor unit recruitment tracings in synchronization with a surface microphone tracing. For spontaneous activity, a sweep speed of 10 msec per division and gain of 50 μV per division was used. For motor unit analysis, sweep speed was the same but gain was changed to 200 μV per division. Patients were not sedated, but received a subcutaneous injection of approximately 0.5 mL of 1% lidocaine with a 1:100,000 dilution over the cricothyroid membrane. After the patient's neck was extended to improve exposure, the concentric needle electrode was inserted and used to locate the thyroarytenoid/lateral cricoarytenoid (TA/LCA) muscle complex. Patients were asked to perform both a sustained vowel /i/ phonation and valsalva maneuver.

Patients with normal or mild to moderate decreased motor recruitment were included in the analysis. We excluded patients with no voluntary motor unit recruitment because they typically do not recover unless there is temporary conduction block.[1] Our methodology of grading motor unit recruitment using an ordinal scale has been previously described,[1] and ranges between 4+ or absent recruitment, to 1+ or mildly decreased activation based on the number of units on the screen and the firing rate of the fastest unit. Synkinesis testing was performed on all patients who demonstrated mild to moderate decreased recruitment (1+ to 2) to assess for normal or misdirected reinnervation of the TA/LCA muscle complex. Synkinesis testing was performed using several trials of sniff maneuvers before and after a sustained /i/ to determine if TA/LCA motor unit amplitude was elevated during the sniff compared to the /i/ maneuver.[3,6] Synkinesis was determined to be present if the ratio of MUP amplitude of sniff/phonation was greater than 0.65; this was based upon our previous research and consistent with our previous evaluation protocol.[3] Turns analysis comprised the mean number of turns/sec, standard deviation of the number of turns/sec, and the number of points outside of the "normal" cloud of several trials of /i/ vocalization at varying intensities in several muscle fields. A normal cloud was created from normative data, as reported previously.[4] All patients underwent electromyography of their normal mobile VF, acting as their own control. These studies were normal, and not involved in further data analysis.

Outcomes and Analysis

Determining the prognosis for recovery of VF motion first takes into consideration the time from injury. Patients were then classified into one of three categories based upon motor recruitment and presence of spontaneous activity (fibrillations, positive waves, and complex repetitive discharges): excellent, fair, or poor, based upon work of Munin et al.[1,7] An excellent prognosis was defined by the presence of normal or near-normal (1+) motor recruitment configuration and the absence of fibrillations or positive wave potentials. A fair prognosis was given when moderately decreased motor unit recruitment (2+ or greater), defined by three or fewer motor units with firing frequency of equal to or greater than 20 Hz, with a diminished interference pattern. A poor prognosis was assigned if motor recruitment was absent (4+) or greatly decreased (3+) defined by only one motor unit firing at equal to or greater than 20 Hz with a severely decreased interference pattern, with or without polyphasic action potentials. Spontaneous activity may be present. Only patients with normal, 1+, or 2+ motor recruitment were analyzed in this study, and therefore were initially given excellent or fair prognosis ratings. Patients with synkinetic activity had their prognosis downgraded regardless of motor recruitment or MU morphology as per our previously described protocol.[3] Turns analysis was then incorporated, with patients with greater than 400 mean turns/sec with improved prognosis, and those with less than 400 mean turns/sec with poorer prognosis. If the patient was bordering between two prognostic categories (e.g., excellent and fair), then turns analysis was subjectively used to determine the prognosis category. All LEMGs were recorded and reviewed to ensure proper classification. Data were analyzed with SPSS for Windows, Release 17.0.0. 2008 (SPSS Inc., Chicago, IL). Means and standard deviations were computed for continuous variables and frequencies were calculated for categoric variables. Relationships between turns, prognosis, and motion recovery were evaluated using Fisher's exact chi-squares.

Assessment of VF Motion

All patients underwent initial and follow-up video recorded flexible laryngoscopy to assess VF motion status. Clinically, unilateral VF immobility had to be present upon presentation with subsequent LEMG confirmation of neuropathy for inclusion in the study. On the contrary, any VF motion that contributed to abduction during respiration and adduction during phonation qualified as "purposeful" VF motion, and these patients were not included in this study.

RESULTS

Twenty-three of the 89 patients met the inclusion criteria. The average age of the study population was

15314995, 2012, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/lary.21884 by University of Dundee, Wiley Online Library on [12/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE I.
Description of Study Population.

Total patients	23
Age (yr), SD	52.7 ± 14.6
Female	73.9%
Duration of symptoms at LEMG (wk), SD	14.0 ± 6.3
Motion recovery, time from onset (wk), SD	14.5 ± 7.2
Idiopathic RLN injury	13.0%
Iatrogenic RLN injury	82.6%
Central vagal injury	4.3%

LEMG = laryngeal electromyography; RLN = recurrent laryngeal nerve; SD = Standard deviation

52.7 ± 14.6 years, and 73.9% were female. An idiopathic injury was suspected in 21.7% of patients, with the remaining suspected to be related to iatrogenic injury. VF motion recovery, verified by flexible laryngoscopy, occurred at an average of 14.5 weeks (range = 3–27 weeks). Paralysis was persistent in all other patients to at least 6 months after onset of symptoms. Demographic and descriptive data are shown in Tables I and II.

LEMG Data

Spontaneous activity. None of the six patients with motion recovery had any spontaneous activity (fibrillations or positive waves). Whereas 8/17 (47%) patients who had no VF motion recovery had fibrillations, with or without associated positive waves.

Recruitment. Five of six patients (83%) with subsequent motion recovery had 1+ decreased motor recruitment, with one patient with normal recruitment. Of the 17 patients with persistent VF paralysis, 1 patient had normal recruitment (6%), 8 had 1+ decreased recruitment (47%), and 8 had 2+ decreased recruitment (47%).

Configuration. Evaluation of the presence of polyphasic action potentials in those patients with return of function (3/6, 50%) was similar to those who did not have return of motion (8/17, 47%). This indicates that prior denervation with subsequent reinnervation is seen in a large percentage of patients with some motor unit recruitment after injury.

Synkinesis. Seven of the 23 total patients (30.4%) were rated to have synkinetic activity based upon a greater than 0.65 ratio of sniff/vocalization. Six of those seven patients (85.7%) had no motion recovery, regardless of MU morphology or degree of recruitment.

Turns analysis. The average mean turns of the six patients with VF motion recovery was 305 turns/sec (SD 122 turns/sec). In contrast, this dropped to 257 turns/sec (SD 110 turns/sec) in the 11 patients with persistent paralysis and no evidence of synkinesis, as those with synkinesis are not expected to recover motion. This difference in mean turns was not significantly different ($P = .68$). Examples of turns analysis data is shown in Figure 1.

When spontaneous activity, motor recruitment configuration, and synkinesis were evaluated together for differences between patients with VF motion recovery and persistent paralysis, a pattern emerged. All patients (6/6) who recovered motion had absence of fibrillations and positive waves, normal or 1+ decreased recruitment, and no detected synkinesis in five of six patients.

Three of the 17 (17.6%) patients who did not recover VF motion also had no fibrillations or positive waves, 1+ recruitment, and no synkinesis. These three patients were rated to have fair and poor prognosis based upon their turns analysis and timing of the LEMG from onset of symptoms. One of these patients showed a neurogenic pattern on turns analysis with mean 570 turns/sec (SD 132 turns/sec) with more than expected points outside the normative cloud (>3) at 27 weeks after onset due to large amplitude measurement. This means that a large percentage of muscle fibers were denervated initially and had been reinnervated by surviving axons. Considering this and the time from onset, further motion recovery was not expected in this patient and therefore was given a "poor" prognosis for further motion recovery. The second patient had slightly decreased mean turns (344 turns/sec, SD 155 turns/sec) 3 weeks after onset, and was given a "fair" prognosis. The third patient had decreased mean turns (249 turns/sec, SD 135 turns/sec) 14 weeks after onset, with a designated "fair" prognosis.

Predictive Values

Based upon qualitative and quantitative data, prognosis for VF motion recovery was assigned to each patient. The predicted value for a positive test (motion recovery) was 100% (4/4) in patients designated with an excellent prognosis. The predicted value of a negative test (no motion recovery) was 89.5% (17/19) in patients given a poor or fair prognosis. No difference is seen between the fair and poor prognosis categories. Negative and positive predictive values, sensitivity, and specificity are shown in Tables III and IV.

DISCUSSION

By using qualitative and quantitative LEMG measurements, our ability to provide accurate prognoses for VF motion recovery in patients with acute RLN injury with some motor recruitment improved. Patients with abnormal but present residual motor unit activity after injury are challenging because qualitative LEMG can only show that some muscle fibers are innervated. This amount of electrical activity is hard to categorize in regard to return of VF motion. Our additional techniques described in this article help to close this

TABLE II.
Iatrogenic RLN Injuries.

Thyroid/parathyroid/central compartment neck dissections	68.4%
Mediastinal/lung	15.8%
Anterior cervical spine surgery	10.5%
Paraganglioma	5.3%

RLN = recurrent laryngeal nerve.

15314995, 2012, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/lary.21884 by University of Dundee, Wiley Online Library on [12/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

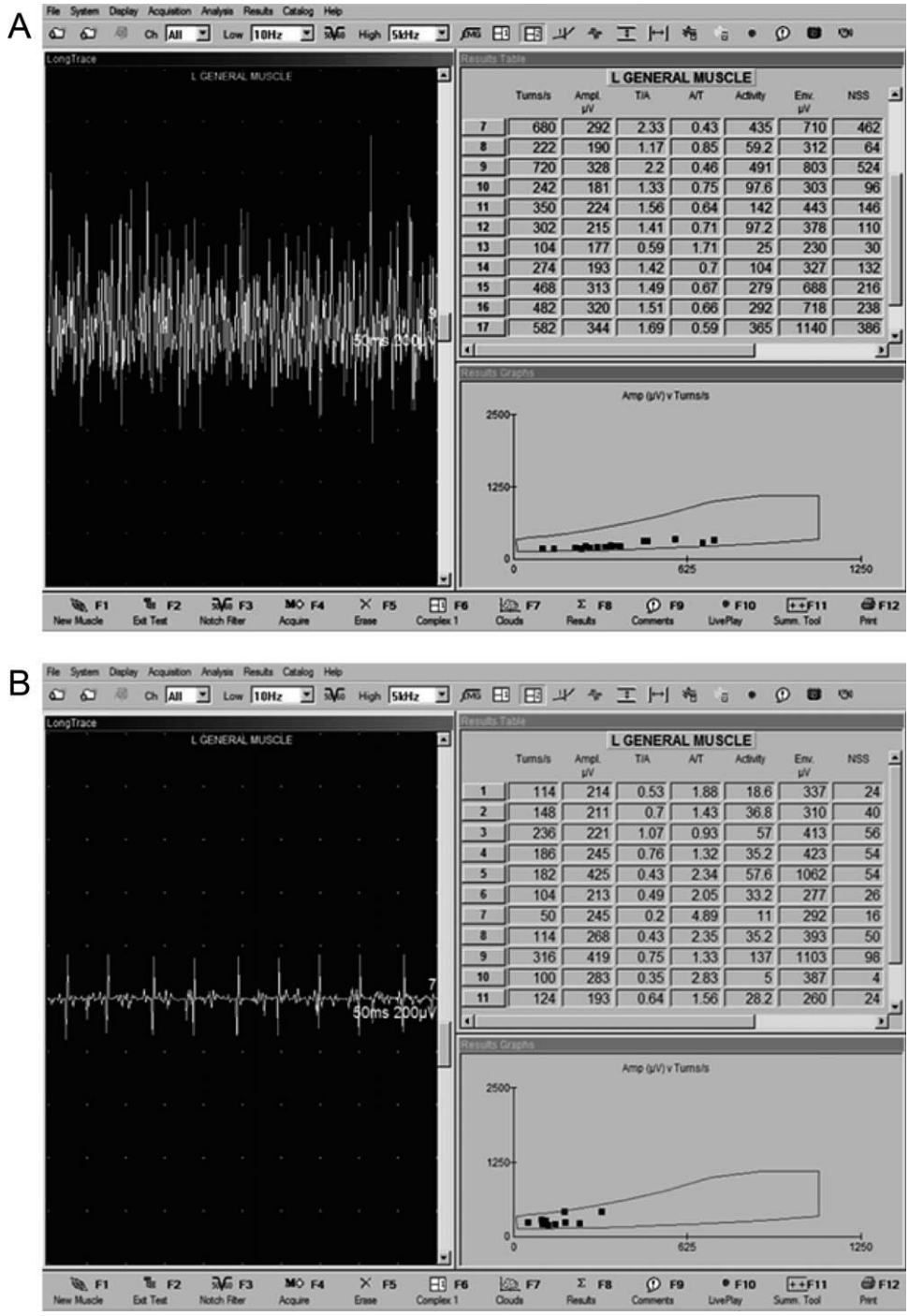

Fig. 1. These are examples of "screen shots" during quantitative LEMG. The left side of the screen shows the EMG tracing. The right upper box demonstrates the number of turns and amplitude for each trial. Several trials are recorded for each patient. The right lower box contains each trial plotted (amplitude vs. turns) within the preestablished "cloud" created from normative data of patients with normal vocal fold motion and no neuropathy. (a) This tracing is of a normal mobile vocal fold with no signs of neuropathy. Note the robust motor recruitment. (b) This tracing is of a patient with vocal fold immobility on flexible laryngoscopy. The patient was found to have no recruitment, 3+ fibrillations, and 2+ positive waves. Note the lack of MU recruitment on the EMG tracing, lower turns/sec, and shift to the left of points within the "cloud." This example of severe neuropathy is for demonstration purposes, as it was not included in analysis for this study, as these patients are uniformly given a "poor" prognosis for VF motion recovery.

knowledge gap. This study responds to the call from the Neurolaryngology Study Group to standardize methods and reporting to improve communication and patient care.[8] Our stepwise approach to determining prognosis by integrating qualitative and quantitative LEMG findings improves predictive values and accuracy.

LEMG Data

Following the progression of LEMG interpretation, Munin et al.[1] reported varying accuracy in predicting motion recovery in the setting of acute VF paralysis based upon the three a priori classifications of motor recruitment. That study showed that two-thirds of patients with excellent prognosis recovered motion, but this diminished to 25% when patients were determined to have fair prognosis, and only 17.6% in patients with poor prognosis. However, the NPV was only 67% and PPV was 80%. Improved accuracy in predicting VF motion recovery occurred with incorporating the presence or absence of synkinesis in patients with good motor recruitment.[3] Hillel[9] elegantly demonstrated concurrent muscle activity of both VF adductors and abductors during rapid respiration or sniff maneuvers in

15314995, 2012, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/lary.21884 by University of Dundee, Wiley Online Library on [12/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE III.
Crosstabulation of Prognosis and Motion Recovery.

		Return of Motion		
		No	Yes	Total
Prognosis	Poor	7 87.5%	1 12.5%	8
	Fair	10 90.9%	1 9.1%	11
	Poor/Fair	17 88.2%	2 11.8%	19
	Excellent	0 0%	4 100%	4

normal subjects. However, when significant pathologic reinnervation occurs between abductor and adductor neurons after nerve injury, VF motion does not recover. Identifying synkinesis can prevent the practitioner from expecting motion recovery in such situations, thus guiding future treatment. The incidence of synkinesis (30%) in our study population was higher than that previously reported.[3,6] We hypothesize this may be due to a higher proportion of iatrogenic surgical injuries leading to greater neurologic injury.

Laryngeal turns analysis was first described in 1990 by Lindestad et al.,[10] with a decrease in turns and amplitude proportional to decreased muscle force. This decrease in muscle force could be from neuropathy, but also from poor patient effort in performing the vocal/valsalva task. Regardless, Statham et al.[4] showed the mean turns analysis could differentiate between acute VF paralysis and normals. Patients tolerate testing well, with the majority of time taken to locate the target muscle. Once the TA/LCA muscle complex has been identified, a complete study is performed in about 10 minutes. The addition of synkinesis and turns methodology only adds a couple minutes of testing time. Mean turns analysis was somewhat helpful based on the small sample of data we have presented. The utility of quantitative measurements is attractive, as it may diminish the subjective nature of LEMG interpretation. In addition, it provides a number that can be tracked over time, possibly providing more information about the state of reinnervation.

We found that VF motion returned in only 12.5% (1/8) of patients with poor prognosis, and 9.1% (1/11) with fair prognosis. Both the "poor" and "fair" prognosis groups had fibrillations (poor 50%, 4/8; fair 37%, 4/11) and synkinesis (poor 50%, 4/8; fair 27%, 3/11) to a greater extent than seen in those with return of VF motion. Thus, it may be more practical to combine fair and poor groups in terms of prognosis, as their rate of VF motion recovery is similar, which is consistent with previous reports.[1]

In this select population of patients with some motor unit activity after injury, we demonstrated a NPV of 89.5% when combining data from both fair and poor prognosis categories, with 66.6% sensitivity, 100% specificity, and 91.3% accuracy. Sensitivity was low, likely due to the low numbers of patients with return of VF motion recovery (4). However, the specificity and accuracy are robust. This suggests more accurate predictions are made with the added information of quantitative LEMG analysis.

Integrating turns analysis into developing a prognosis algorithm is still being explored and studied. Using both qualitative and quantitative LEMG measurements, our ability to accurately predict VF motion recovery in the setting of acute VF paralysis and some motor recruitment has improved. Regression analysis of each component's relative weight (timing from onset, MU morphology, recruitment, synkinesis, and turns analysis) in determining prognosis for recovery remains unclear and further studies to elucidate this are necessary.

Limitations

Acute VF paralysis was attributed to an iatrogenic etiology 82.6% of the time in our study. This is somewhat high, and reflects referral patterns at our tertiary care institution. This may have contributed to a greater incidence of synkinesis in our study when compared with other reports. Although LEMG findings have been found to be more important in reliably predicting VF motion recovery than etiology of the lesion,[11] iatrogenic and idiopathic etiologies may need to be evaluated differently. Even within iatrogenic neurogenic injuries, studies suggest that endotracheal intubation as an etiology for VF paralysis has a more favorable natural recovery.[12,13] Large prospective natural history studies are needed to determine the impact of different etiologies on VF motion recovery.

The average time from onset of VF paralysis and LEMG was variable in our study, with an average of 14.5 weeks. The timing of LEMG is critical in determining prognosis for VF motion recovery. However, the natural history of VF paralysis associated with different etiologies and LEMG findings is unknown at this time.

TABLE IV.
Predictive Values of Motion Recovery.

NPV	PPV	Sensitivity	Specificity	Accuracy
TN/(TN+FN)	TP/(TP+FP)	TP/(TP+FN)	TN/(TN+FP)	(TP+TN)/(TP+TN+FP+FN)
89.5%	100%	66.6%	100%	91.3%

NPV = negative predictive value; PPV = positive predictive value; TN = true negative; FN = false negative; TP = true positive; FP = false positive.

15314995, 2012, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/lary.21884 by University of Dundee, Wiley Online Library on [12/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Therefore, the optimal time to obtain LEMG information is yet to be determined. Several of our patients had two serial LEMGs separated by 6 to 8 weeks. Only the LEMG closest to 2 months was analyzed per patient, based upon previous studies.[1,2,14] Serial LEMGs may improve predictability and therefore definitive treatment,[11] but further research is needed.

Rare inadvertent misplacement of the needle electrode within the strap muscles occurs, usually associated with challenging patient anatomy or recent anterior cervical surgery (e.g., thyroidectomy). In these cases, the thyroid cartilage is identified with the needle electrode, and slowly "walked" down to the cricothyroid membrane. If the needle is properly position, there should be no recruitment with neck flexion against resistance. These "tricks" help minimize inadvertent misplacement of the needle electrode.

If the patient has significant pain with the procedure, the MU recruitment and amplitude may be artificially attenuated. This must be noted and the results evaluated with that in mind.

When performing synkinesis testing, the quality of sniff maneuver is important. If the sniff is too vigorous, artifact occurs and obscures the actual presence or absence of synkinetic activity. Several trials are often necessary, with coaching of the patient on how to correctly perform a sniff. Because the presence of synkinesis is critical for downgrading prognosis of VF motion recovery, false positives may lead to inappropriate treatment.

CONCLUSION

This study showed an improvement in accurately predicting motion recovery in patients with acute VF paralysis in the presence of normal or moderately decreased motor recruitment with evaluation of MU characteristics, presence of synkinesis, and quantitative mean turns analysis. Determining a potential role for quantitative laryngeal electromyographic analysis is appealing, as it could diminish the necessity of relying on subjective ratings of MU morphology and recruitment. This study is a first step toward that goal. As our understanding of LEMG advances, our ability to properly counsel patients also improves.

BIBLIOGRAPHY

1. Munin MC, Rosen CA, Zullo T. Utility of laryngeal electromyography in predicting recovery after vocal fold paralysis. *Arch Phys Med Rehabil* 2003;84:1150–1153.
2. Wang CC, Chang MH, Wang CP, Liu SA. Prognostic indicators of unilateral vocal fold paralysis. *Arch Otolaryngol Head Neck Surg* 2008;134: 380–388.
3. Statham MM, Rosen CA, Smith LJ, Munin MC. Electromyographic laryngeal synkinesis alters prognosis in vocal fold paralysis. *Laryngoscope* 2010;120:285–290.
4. Statham MM, Rosen CA, Nandedkar SD, Munin MC. Quantitative laryngeal electromyography: turns and amplitude analysis. *Laryngoscope* 2010;120:2036–2041.
5. Woodson GE. Spontaneous laryngeal reinnervation after recurrent laryngeal or vagus nerve injury. *Ann Otol Rhinol Laryngol*;1:57–65.
6. Maronian NC, Robinson L, Waugh P, Hillel AD. A new electromyographic definition of laryngeal synkinesis. *Ann Otol Rhinol Laryngol* 2004;13: 877–886.
7. Munin MC, Rainer M. laryngeal electromyography. In: Carrau RL, Murry T, eds. *Comprehensive Management of Swallowing Disorders*. San Diego, CA: Singular Publishing Group; 1999:87–90.
8. Blitzer A, Crumley RL, Dailey SH, et al. Recommendations of the Neurolaryngology Study Group on laryngeal electromyography. *Otolaryngol Head Neck Surg* 2009;140:782–793.
9. Hillel AD. The study of laryngeal muscle activity in normal human subjects and in patients with laryngeal dystonia using multiple fine-wire electromyography. *Laryngoscope* 2001;111(Pt 2, Suppl 97):1–47.
10. Lindestad PA, Fritzell B, Persson A. Evaluation of laryngeal muscle function by quantitative analysis of the EMG interference pattern. *Acta Otolaryngol* 1990;109:467–472.
11. Grosheva M, Wittekindt C, Pototschnig C, Lindenthaler W, Guntinas-Lichius O. Evaluation of peripheral vocal cord paralysis by electromyography. *Laryngoscope* 2008;118:987–990.
12. Yamada M, Hirano M, Ohkubo H. Recurrent laryngeal nerve paralysis. A 10-year review of 564 patients. *Auris Nasus Larynx* 1983;10(Suppl): S1–S15.
13. Santos PM, Afrassiabi A, Weymuller EA Jr. Risk Factors associated with prolonged intubation and laryngeal injury. *Otolaryngol Head Neck Surg* 1994;111:453–459.
14. Min YB, Finnegan EM, Hoffman, HT, Luschei ES, McCulloch TM. A preliminary study of the prognostic role of electromyography in laryngeal paralysis. *Otolaryngol Head Neck Surg* 1994;111:770–775.

15314995, 2012, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/lary.21884 by University of Dundee, Wiley Online Library on [12/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License